

SHANU KUMAR MANGLAM

Chittaranjan, West Bengal | +91 7903399754 | shanukr44150@gmail.com

Portfolio: Shanu Kumar Manglam — GenAI Engineer

github.com/Shanu1110 | linkedin.com/in/shanu1110

Professional Summary

AI/ML Software Engineer specializing in Generative AI, AI Agents, LLMs, and RAG pipelines. Experienced in orchestrating multi-agent workflows, prompt engineering, and building Python backend APIs using Google ADK, LangChain, FAISS, and Microsoft Copilot Studio. Skilled in developing cloud-native microservices with Docker, Google Cloud Run, and CI/CD automation.

Technical Skills

- AI Agents & Frameworks:** Google ADK, Microsoft Copilot Studio, LangChain, Agentic Workflows, Tool Calling, Multi-Agent Orchestration
- Generative AI & RAG:** Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), Embedding Models, Semantic Search, Prompt Engineering, LLMs (GPT-4o, Gemini API), Responsible AI
- AI/ML & Computer Vision:** Machine Learning, Deep Learning, NLP, OpenCV, PyTorch, TensorFlow, YOLO, Feature Engineering, Model Validation
- Backend & Cloud DevOps:** Python, RESTful APIs, FastAPI/Streamlit, Docker, Google Cloud Run, GCP, AWS, Microservices Architecture, Git, GitHub Actions, CI/CD, Playwright
- Data Engineering & Analytics:** SQL, PySpark, ETL Pipelines, EDA, Data Cleansing, Data Quality, Power BI, Looker Studio
- Methodologies & SDLC:** Agile, Scrum, Code Reviews, Technical Documentation, Testing, Debugging, End-to-End AI Lifecycle

Professional Experience

Cognizant

Chennai, India

Programmer Analyst Trainee

Feb 2026 – Apr 2026

- Engineered AI-driven agentic workflows, document summarization, and reporting using Microsoft Copilot Studio and prompt engineering with strict Responsible AI guardrails.
- Built scalable ETL pipelines in Python and PySpark to cleanse, transform, and validate large structured marketing datasets for analytics and AI models.
- Developed analytics dashboards integrating GA4, GTM, Adobe Analytics, and Looker Studio to track campaign KPIs in an Agile environment.

Ernst & Young

Remote

Data Analyst Intern

Feb 2024 – Apr 2024

- Developed predictive analytics models and interactive Power BI KPI dashboards using Python and SQL to support healthcare data decisions.
- Executed end-to-end EDA and ETL pipelines to cleanse and transform structured datasets, boosting reporting accuracy by 20%.

Projects

Cloud-Native RAG AI Agent & Orchestration | *Google ADK, Google Cloud Run, Streamlit, Python* Jul 2026 – Aug 2026

- Built and deployed an end-to-end RAG AI Agent using Google ADK (Agent Development Kit) integrated with Streamlit and containerized on Google Cloud Run.
- Designed ADK 2 Orchestration workflows implementing graph-based, collaborative, and dynamic agent architectures with built-in tools, instructions, and reasoning chains.
- Implemented cloud-native microservices deployment principles, containerizing the application using Docker and establishing reliable cloud execution.

HR Policy Assistant – Enterprise RAG Application | *FAISS, LangChain, GPT-4o, Docker, Streamlit, REST APIs* May 2026

- Engineered an enterprise Retrieval-Augmented Generation (RAG) assistant using LangChain, GPT-4o, FAISS vector database, and Streamlit for context-aware document question answering.

- Built data ingestion, text chunking, tokenization, vector embeddings, and semantic search retrieval pipelines, improving retrieval relevance by approximately 35%.
- Configured prompt engineering, instruction sets, and response evaluation guardrails to ensure factual accuracy and Responsible AI compliance.
- Developed backend REST API endpoints, containerized the full service with Docker, and configured automated CI/CD deployment pipelines using GitHub Actions.

ATS Resume Copilot – Agentic Job Search Automation | *Gemini API, Playwright, LangChain, Streamlit* Apr 2026

- Designed an autonomous agentic platform integrating Gemini API, Playwright, LangChain, and Streamlit to automate multi-step job search workflows.
- Automated resume-to-JD parsing, ATS score optimization, web scraping, and application tracking, reducing manual effort by over 50%.
- Implemented workflow automation, authentication handling, browser automation tools, and API integrations for scalable operations.
- Refined agent instructions and LLM outputs through prompt optimization, structured evaluation, and systematic debugging.

Research Paper Retrieval System | *Transformers, FAISS, Embedding Models, Streamlit* Mar 2026

- Built an intelligent academic search engine using Retrieval-Augmented Generation, FAISS vector indexing, transformers, and embedding models.
- Optimized semantic retrieval pipelines, chunking strategies, metadata filtering, and document ranking for high-accuracy document search.
- Benchmarked retrieval workflows to maximize latency performance and output relevance across large research datasets.

Education

University of Engineering & Management, Kolkata
Bachelor of Technology in Electronics and Communication Engineering (CGPA: 8.86 / 10)

Kolkata, India
Aug 2021 – Jul 2025

Awards & Honors

Vice Chancellor’s Award for Best Scientific Mind 2025
Focus: Artificial Intelligence, Machine Learning, Electronics Engineering

May 2025

- Awarded for outstanding innovation and research contributions in AI, Machine Learning, and Electronics Engineering.

Publications

Review Paper: Renewable Energy Trends for the Next Decade
UEMGREEN’25

Apr 2025

- Published a review on AI-driven renewable energy technologies, grid modernization, and sustainable energy systems.

Research Paper: Analysing CO2 Emission from Vehicles Using Machine Learning
UEMGREEN’25

Apr 2025

- Developed, trained, and benchmarked Machine Learning models for predicting vehicular CO2 emissions to support sustainability analytics.

Certifications

- **Generative AI & Agents:** Deploying RAG AI Agents with Google ADK + Cloud Run (Google Developer Program), ADK 2 Orchestration: Graph, Collaborative & Dynamic Workflows, Agentic AI Mastery, Data & AI Foundations for ATCI, Generative AI for Intermediate, Prompt Engineering, Context Engineering, Responsible AI
- **Developer Tools:** Microsoft Copilot Studio, GitHub Copilot Complete Guide, Apache Spark 3 – Data Engineering
- **Cloud & Analytics:** AWS Cloud Fundamentals, Google Analytics, GA4, Google Tag Manager, IBM Data Science Professional Certificate